

1.



SIMATS Online Programming Lab

C++ PROGRAMMING

SHRUTHI SREE R
192421129RC

QUESTIONS 9 / 9 completed

GCD-Fun.

GCD-FUN.

Write a function to determine the GCD (greatest common divisor) of two given integers.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 // Function to calculate the Greatest Common Divisor (GCD) using Euclidean Algorithm
5 int findGCD(int a, int b) {
6     while (b != 0) {
7         int temp = b;
8         b = a % b;
9         a = temp;
10    }
11    return a;
12 }
13
14 int main() {
15     int a, b;
16     cin >> a >> b;
17
18     cout << findGCD(a, b) << endl;
19
20     return 0;
21 }
```

INPUT

56 98

eeetha.com/practice_app/dashboard_student

2.



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QUESTIONS 9 / 9 completed

Reverse

REVERSE

Write a function to reverse a given string.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <string>
3 #include <algorithm>
4 using namespace std;
5
6 // Function to reverse a given string
7 string reverseString(string str) {
8     reverse(str.begin(), str.end());
9     return str;
10 }
11
12 int main() {
13     string input;
14     cin >> input;
15
16     cout << reverseString(input) << endl;
17
18     return 0;
19 }
```

INPUT

Hello

3.



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QUESTIONS 9 / 9 completed

Count

COUNT
Write a function to count the number of words in a given string.

PROGRAM EDITOR
C++
Run Save

```
1 #include <iostream>
2 #include <string>
3 #include <sstream>
4 using namespace std;
5
6 // Function to count the number of words in a given string
7 int countWords(const string& str) {
8     stringstream ss(str);
9     string word;
10    int count = 0;
11
12    while (ss >> word) {
13        count++;
14    }
15
16    return count;
17 }
18
19 int main() {
20     string input;
21     getline(cin, input);
```

INPUT
Hello world welcome
to SIMATS
programming

4.



SIMATS Online Programming Lab
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QUESTIONS 9 / 9 completed

Min-Max

MIN-MAX
Write a function to find the minimum and maximum elements in a given array.

PROGRAM EDITOR
C++
Run Save

```
1 #include <iostream>
2 using namespace std;
3
4 // Function to find minimum and maximum in an array
5 void findMinMax(int arr[], int n, int &minVal, int &maxVal) {
6     minVal = arr[0];
7     maxVal = arr[0];
8
9     for (int i = 1; i < n; i++) {
10        if (arr[i] < minVal) {
11            minVal = arr[i];
12        }
13        if (arr[i] > maxVal) {
14            maxVal = arr[i];
15        }
16    }
17 }
18
19 int main() {
20     int n;
21     cin >> n;
```

INPUT
5
12 45 2 67 23

5.

QUESTIONS 9 / 9 completed

Palindrome-Fun. ▾

PALINDROME-FUN.

Write a function to check if a given string is a palindrome or not.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 #include <string>
3 #include <algorithm>
4 using namespace std;
5
6 // Function to check if a string is a palindrome
7 bool isPalindrome(const string& str) {
8     string rev = str;
9     reverse(rev.begin(), rev.end());
10    return str == rev;
11 }
12
13 int main() {
14     string str;
15     cin >> str;
16
17     if (isPalindrome(str)) {
18         cout << str << " is a palindrome." << endl;
19     } else {
20         cout << str << " is not a palindrome." << endl;
21     }
22 }
```

INPUT

madam